

ABHISHEK DAS

Senior Applied Scientist | Multimodal ML | Retrieval & Ranking

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Summary

Senior Applied Scientist building production multimodal search, retrieval, ranking, and low-latency inference systems at Microsoft. Led development of vision-language models, approximate nearest-neighbor retrieval, and bandit systems at billion-query scale, driving measurable engagement and revenue impact.

Experience

Microsoft AI

Redmond, WA

Senior Applied Scientist

08/2021 - Present

- Designed and productionized a low-latency vision-language model (VLM) powering multimodal autosuggest in Bing Visual Search, achieving **sub-100 ms p95 latency** through layer pruning and inference optimization.
- Led VLM-powered ads query generation and Thompson Sampling-based bandit ranking for Bing Image Search, driving an **estimated \$10M in annual revenue** and increasing Product Ads clicks per user by **9.4%**.
- Developed and shipped online approximate nearest-neighbor recall for ads query selection, generating and publishing a **billion-scale query index** and integrating the recall path into production serving, driving an **estimated \$2M in annual revenue**.
- Modernized Related Searches across Bing SERP and Image Search with deep-learning recall and ranking, deploying distilled student transformers at billion-query scale; increased clicks by **over 10%**, coverage by **up to 50%**, and enabled weekly refreshes across 14 languages.
- Developed Bing Image Creator's first neural prompt-safety model and evaluation framework for sensitive content, reducing overblocking and harmful-generation leakage for a product serving **approximately 2M daily active users**.
- Promoted three times** since joining Microsoft; now lead technical delivery across model development, online experimentation, Responsible AI approval, and production deployment while mentoring junior scientists and reviewing internal research.

Carnegie Mellon University

Pittsburgh, PA

Deep Learning Researcher

05/2020 - 07/2021

- Developed fine-grained retail product recognition models for noisy, long-tail datasets, applying CutMix, label smoothing, attention mechanisms, and targeted sampling strategies to production data.
- Co-invented a deep-learning shelf-management system for automated inventory monitoring, granted as U.S. Patent US12437258B2.

Education

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Electrical and Computer Engineering

08/2019 - 12/2020

University of Mumbai

Mumbai, India

Bachelor of Engineering in Electronics and Telecommunication Engineering

08/2015 - 05/2019

Selected Research & Patents

Detecting Hate Speech in Multi-modal Memes - Facebook Hateful Memes Challenge, NeurIPS 2020 Contributed Talk | **110+** citations | [arXiv](#)

Multi-Image Steganography Using Deep Neural Networks | **60+** citations | [arXiv](#), 2021

System and Method for Identifying Products in a Shelf Management System - [U.S. Patent](#) US12437258B2, 2025 | **30+** citing patent families

Skills

Methods: Vision-Language Models • Multimodal Retrieval • Preference Optimization • Knowledge Distillation • Bandits

Languages & Tools: Python • PyTorch • Hugging Face Transformers • vLLM • ONNX Runtime • DeepSpeed • PEFT • FAISS • CUDA • Linux